

Name :-_Pratiksha Madhav Tonde

Clg Name :- T.B.Girwalkr Polytechnic , Ambajogai

Domain :- AI - ML

Branch Name :- Computer Engineering

1)

```
*** (2500,)
```

2)

```
*** Index(['Brand', 'Fuel Type', 'Transmission', 'Condition', 'Model'], dtype='object')
   Car ID  Year  Engine Size  Mileage  Brand_BMW  Brand_Ford  Brand_Honda  \
0        1  2016          2   114832          0          0          0
1        2  2018          4   143190          1          0          0
2        3  2013          4   181601          0          0          0
3        4  2011          4    68682          0          0          0
4        5  2009          2   223009          0          1          0

   Brand_Mercedes  Brand_Tesla  Brand_Toyota  ...  Model_Model S  \
0                0            1            0  ...              0
1                0            0            0  ...              0
2                0            0            0  ...              0
3                0            1            0  ...              0
4                0            0            0  ...              0

   Model_Model X  Model_Model Y  Model_Mustang  Model_Prius  Model_Q5  \
0                1            0            0            0            0
1                0            0            0            0            0
2                0            0            0            0            0
3                0            1            0            0            0
4                0            0            1            0            0

   Model_Q7  Model_RAV4  Model_X3  Model_X5
0          0          0          0          0
1          0          0          0          0
2          0          0          0          0
3          0          0          0          0
4          0          0          0          0

[5 rows x 43 columns]
```

3)

```

***   Car ID      Year  Engine Size  Mileage  Brand_BMW  Brand_Ford  \
0      1  0.340641  -0.449538 -0.214571         0         0
1      2  0.343955   0.778036  0.083988         1         0
2      3  0.335670   0.778036  0.488387         0         0
3      4  0.332356   0.778036 -0.700447         0         0
4      5  0.329042  -0.449538  0.924338         0         1

      Brand_Honda  Brand_Mercedes  Brand_Tesla  Brand_Toyota  ...  Model_Model S  \
0              0              0              1              0  ...              0
1              0              0              0              0  ...              0
2              0              0              0              0  ...              0
3              0              0              1              0  ...              0
4              0              0              0              0  ...              0

      Model_Model X  Model_Model Y  Model_Mustang  Model_Prius  Model_Q5  \
0              1              0              0              0              0
1              0              0              0              0              0
2              0              0              0              0              0
3              0              1              0              0              0
4              0              0              1              0              0

      Model_Q7  Model_RAV4  Model_X3  Model_X5
0              0              0              0              0
1              0              0              0              0
2              0              0              0              0
3              0              0              0              0
4              0              0              0              0

[5 rows x 43 columns]

```

4)

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***  X_train shape: (1675, 43)
     X_test shape : (825, 43)
     y_train shape: (1675,)
     y_test shape : (825,)

```

5)

	Feature	Coefficient
0	Car ID	0.872265
1	Year	-78488.686773
2	Engine Size	-162.898602
3	Mileage	-562.651069
4	Brand_BMW	5117.409382
5	Brand_Ford	-1106.439448
6	Brand_Honda	-1022.129831
7	Brand_Mercedes	-473.603451
8	Brand_Tesla	1086.730293
9	Brand_Toyota	-97.419488
10	Fuel Type_Electric	-2866.044530
11	Fuel Type_Hybrid	-2040.303671
12	Fuel Type_Petrol	-3316.132240
13	Transmission_Manual	1232.547422
14	Condition_New	-691.340206
15	Condition_Used	-216.385406
16	Model_5 Series	-2944.151869
17	Model_A3	-502.447863
18	Model_A4	-1567.374225
19	Model_Accord	2797.215878
20	Model_C-Class	-368.642604
21	Model_CR-V	-6252.414155
22	Model_Camry	-1183.284352
23	Model_Civic	747.090785
24	Model_Corolla	3543.537392
25	Model_E-Class	-437.596522
26	Model_Explorer	3218.415965
27	Model_Fiesta	-832.801632
28	Model_Fit	1685.977662
29	Model_Focus	5726.648970
30	Model_GLA	-6005.508856
31	Model_GLC	6338.144530
32	Model_Model 3	-1711.028690
33	Model_Model S	879.803690
34	Model_Model X	-740.986111
35	Model_Model Y	2658.941405
36	Model_Mustang	-9218.702750
37	Model_Prius	-1873.048365
38	Model_Q5	682.288840
39	Model_Q7	-2117.014209
40	Model_RAV4	-584.624163

34	Model_Model X	-740.986111
35	Model_Model Y	2658.941405
36	Model_Mustang	-9218.702750
37	Model_Prius	-1873.048365
38	Model_Q5	682.288840
39	Model_Q7	-2117.014209
40	Model_RAV4	-584.624163
41	Model_X3	-6104.653485
42	Model_X5	-6871.299605

6)

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*** [316126.8532227 316126.8532227 316126.8532227 316126.8532227
316126.8532227 316126.8532227 316126.8532227 316126.8532227
316126.8532227 316126.8532227]
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	Actual	Predicted
0	NaN	316126.853223
1	NaN	316126.853223
2	NaN	316126.853223
3	NaN	316126.853223
4	NaN	316126.853223
5	NaN	316126.853223
6	NaN	316126.853223
7	NaN	316126.853223
8	NaN	316126.853223
9	NaN	316126.853223